Vikram Singh, PhD

MK Bhan Young Reserach Fellow

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Research Interests

My core research lies at the intersection of Systems Biology and Computational Network Science. I focus on identifying higher-order organizational patterns in biological systems — structures like multi-node motifs and dynamic modules that shape biological functions or dysfunctions. My current focus is on integrating multi-omics temporal lobe epilepsy datasets, where I apply hypergraph theory and tensor decomposition to understand how disrupted multi-way interactions drive complex diseases from their molecular origins to observable symptoms.

Education

Central University of Himachal Pradesh — Computational Biology and Bioinformatics
PhD — 2016-2023

Central University of Himachal Pradesh — Computational Biology and Bioinformatics
MSc — 2011- 2013

Himachal Pradesh University — Medical
BSc — 2008-2011

Research Experience

International Centre for Genetic Engineering and Biotechnology — Delhi, India
MK Bhan Young Researcher Fellow — Oct 2024 - Present

- Principal Investigator (MK Bhan Young Researcher Fellowship) of a*Department of Biotechnology (DBT) funded research project* integrating multi-omics data and deep learning to identify phytochemical-based therapeutic candidates for temporal lobe epilepsy.
- Develop hypergraphs by integrating multi-omics data (bulk and single-cell RNA-seq, PPI, epigenomic, and GWAS) to analyze higher-order network topology, stability, and disease-specific rewiring between TLE and non-TLE conditions, and identify system-level regulators of temporal lobe epilepsy.
- Prioritize anti-TLE phytochemicals via target–compound interaction networks and deep learning.

Indian Biological Data Centre, Regional Centre for Biotechnology — Faridabad
Data Curator — Mar 2024 - Sep 2024

- Part of the core development team for the Indian Nucleotide Data Archive (INDA) and its controlled-access version INDA-CA, where I contributed to database design, data workflows, metadata standards, and submission interfaces for secure and regulated sharing of sensitive human genomic data.
- Involved in the developement of data management pipelines for data coming from national projects INSACOG and GenomeIndia, ensuring high-quality, secure handling of large-scale sequencing datasets.
- Processed and analyzed massive whole genome sequencing (WGS) and metagenomic samples (sewage surveillance) datasets from multiple sequencing platforms (Illumina, Ion torrent and Oxford Nanopore technologies). Performed variant calling, annotation, QC, filtering, lineage assignment, and mutation reporting for SARS-CoV-2.

Central University of Himachal Pradesh — Shahpur, Himachal Pradesh
PhD Student (CSIR-JRF and CSIR-SRF)) — Jan 2016 - Sep 2023

- Conducted interdisciplinary research in Systems Biology and Computational Network Science, focusing on higher-order organizational patterns in complex biological systems.
- Analyzed diverse biological and neurological networks, including protein–protein interaction networks, residue-level protein structural graphs, disease gene interaction networks (neoplasms, neurological disorders), and brain functional networks (ASD, ADHD fMRI).
- Employed higher-order network motifs and graphlet orbit profiles to capture multi-node structural patterns overlooked by traditional pairwise interaction models.
- Developed deep learning models trained on graphlet-based features, achieving superior performance over Node2Vec and other state-of-the-art network embedding methods for disease and tissue classification.
- Designed and implemented two novel scale-free null models (density-dependent and stochastic) to simulate biologically realistic constraints such as network density, noise, and heterogeneity, ensuring statistical robustness of network comparisons.
- Built scalable pipelines to construct genome- and transcriptome-wide interaction networks from scratch by integrating orthologous PPI data and tissue-specific transcriptomic profiles.

Sher-e-Kashmir University of Agricultural Science and Technology — Jammu
Junior Research Fellow — Dec 2013 - Mar 2015

- Worked as a Junior Research Fellow on a project focused on developing a high-density SNP genotyping chip for *Brassica juncea*.Performed genomic data processing, alignment, variant calling, and downstream analyses to identify high-confidence, informative SNP markers.
- Performed quality filtering, annotation, and prioritization of SNPs for inclusion in the final chip design.Participated in variant discovery and marker selection using whole-genome resequencing data from multiple *B. juncea* accessions.

Honors and Awards

MK Bhan Young Researcher Fellowship (MKBYRF) May 2024

Awarded the MKBYRF for the project "Unveiling therapeutic potential of phytochemicals against temporal lobe epilepsy (TLE): A multi-omics heterogenous hypergraphs-based deep learning framework" by Department of Biotechnology (DBT), Govt. of India

Department of Biotechnology Research Associateship (DBT-RA) Apr 2024

Awarded the DBT-RAship for the project "Deciphering Genomic and Translational Alterations Driving Drug Resistance in Indian Clinical Isolates of Candida auris: Insights into Antifungal Therapeutic Strategies" by Department of Biotehcnology (DBT), Govt. of india

Himachal Pradesh State Eligibility Test (HP-SET) Oct 2018

Qualified the HP-SET in LIFE SCIENCES conducted by Himachal Pradesh Public Service Commission (HPPSC), (State-wide rank: 20)

CSIR-NET JRF May 2018

Qualified, joint CSIR-UGC National Eligibility Test (NET) and awarded CSIR Junior Research Fellowship (JRF) in LIFE SCIENCES, conducted by the Council of Scientific and Industrial Research. (AIR: 121)

The Research Entrance Aptitude Test 2015

Qualified TREAT for admission to the PhD degree course in Computational Biology and Bioinformatics, conducted by Central University of Himachal Pradesh (CUHP) (AIR: 1)

Further Education Admission Test (FEAT) 2011

Qualified FEAT for admission toMSc Computational Biology and Bioinformatics, conducted by Central University of Himachal Pradesh (AIR: 16)

Publications

Refereed Journal Articles

1. Shah S., Sharma, A., Choudhary, K., Kumar, R., Singh, V., Sharma, AK., Kumar, S. Sharma, D. (2025) Exploring OmpA of Orientia tsutsugamushi to design novel multi epitope vaccine against scrub typhus: an immunoinformatics approach. Molecular Diversity, 1-16. ISSN 1573-501X.
2. Singh, V., & Singh, V. (2023). Characterizing circadian connectome of O. tenuiflorum using an integrated network theoretic framework. Scientific Reports, 13, 13108. ISSN 2045-2322.
3. Singh, G., Singh, V.†, & Singh, V. (2022). Systems scale characterization of circadian rhythm pathway in Camellia sinensis. Computational and Structural Biotechnology Journal, 20, 598-607. ISSN 2001-0370.
4. Singh, V., & Singh, V. (2022). Construction and stochastic scale-free modelling of empirical, global, index-case SARS-CoV-2 transmission network. Journal of Complex Networks, 10(1), 1-15. ISSN 2051-1329.
5. Singh, G., Singh, V.†, & Singh, V. (2021). Genome-wide interologous interactome map (TeaGPIN) of Camellia sinensis. Genomics, 113(1), 553-564. ISSN 0888-7543.
6. Singh, V., Singh, G., & Singh, V. (2019). TulsiPIN: an interologous protein interactome of *Ocimum tenuiflorum*. Journal of Proteome Research, 19(2), 884-899. ISSN 1535-3893.
7. Singh, G., Singh, V., & Singh, V. (2019). Construction and analysis of an interologous protein–protein interaction network of *Camellia sinensis* leaf (TeaLIPIN) from RNA–Seq data sets. Plant Cell Reports, 38(10), 1249-1262. ISSN 0721-7714.

† Contributed equally

Refereed Book Chapters

1. Singh, V., & Singh, V. (2024). “Inferring interaction networks from transcriptomic data: methods and applications” chapter in “Transcriptome Data Analysis”. Eds Azad, R.K. Methods in Molecular Biology, vol 2812. Humana Press, New York, NY: Springer US, ISBN: 9781071638866. Pp. 11-37.
2. Singh, V., and Singh. V. (2023). “Higher order organization in biological systems: an introduction” chapter in “Bioinformatics and Computational Biology: Technological Advancements, Applications and Opportunities”. Eds Tiratha Raj Singh, Hemraj Saini and Moacyr Comar Junior, CRC Press LLC. Florida, ISBN: 9781032361581.
3. Singh, V., and Singh. V. (2014). “Introduction to complex biological networks” chapter in “Biotechnology (Vol. 4): Applied Synthetic Biology”. Eds Vijai Singh and J. N. Govil, Stadium Press LLC. Houston, ISBN: 1-62699-019-0. Pp. 43-88.

In Communication

1. Singh, V., & Singh, V. DeepAutoPIN: An automorphism orbits based deep neural network for characterizing the organizational diversity of protein interactomes across the tree of life. arXiv preprint arXiv:2203.00999v2. DOI: 10.48550/arXiv.2203.00999.
2. Singh, V., & Singh, V. Higher order organizational features can distinguish protein interaction networks of disease classes: a case study of neoplasms and neurological diseases. arXiv preprint arXiv:2212.13171. DOI: 10.48550/arXiv.2212.13171.

Teaching and Scientific Mentoring

Assisted my mentor in teaching core computational Biology courses during PhD 2016 - 2020

Algorithms in computational biology

Basics of sequence analysis

Elements of Systems Biology
Practical course on Perl

Assited my mentor in mentoring 6 MSc students' projects during my PhD 2017-2021

Software, Algorithms & Databases Developed

Tulsi Protein Interaction Network (TulsiPIN) https://doi.org/10.1021/acs.jproteome.9b00683
https://doi.org/10.1 021/acs.jproteome. 9b00683

Density-Dependent Scale-Free (DDSF) Random Network Algorithm
https://github.com/v ikramsinghlab/DDSF

Stochastic Scale-Free (SSF) Random Network Algorithm https://doi.org/10.1093/comnet/cnab047
https://github.com/v ikram0691/C19- TraNet

Network Topology–Based Prediction of Core Circadian Clock Genes https://doi.org/10.1101/2022.03.02.482599
https://github.com/v ikramsinghlab/Pcrp

DeepAutoPIN https://doi.org/10.48550/arXiv.2203.00999
https://github.com/vikramsinghlab/DeepAutoPIN.git

Skills

Coding
Python, Perl, R, Shell Scripting, PhP, Javascript, SQL

Research and Academic
Academic research, algorithm development, teaching, conference/journal publishing.

Languages
Strong reading, writing and speaking competencies for English and Hindi

Conferences and Workshps

Member, Organizing Committee Nov 11-15, 2025
International workshop on “Application of Artificial Intelligence (AI) to Various Healthcare Problems” held at the International Centre for Genetic Engineering and Biotechnology (ICGEB).

Conference poster presentation Dec 3-5, 2022
Presented a poster entitled "Characterizing the Circadian Connectome of a Plant Using the Higher Order Organizational Structures Underlying its Protein Interaction Network," at the "International Conference on Emerging Trends in Biosciences and Chemical Technology-2022 (ETBCT-2022)" organized by Shri Mata Vaishno Devi University Katra-182320 Jammu and Kashmir, India

Conference poster presentation Jul 10-14, 2022
Poster a poster entitled "Automorphism orbits based characterization of protein interaction networks across the three domains of life" at the international conference "Intelligent Systems for Molecular Biology (ISMB-2022)" hosted by the International Society for Computational Biology (ISCB) in Madison, Wisconsin, USA.

Participant Dec 24-30, 2014
Hands on training of implementing wet lab techniques like DNA/RNA extraction. PCR protein extraction, Electrophoresis, Western blotting and other molecular biology techniques, held at Sher-e-Kashmir University of Agricultural Science and Technology, Jammu.

Declaration

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I hereby declare the above details are true to the best of my knowledge and that I’ll do my best for the good of the organization

References

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